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MARKET BASKET ANALYSIS USING APRIORI AND FP-GROWTH ALGORITHMS

Abstract

The objective of this study is to examine customer purchasing behavior in a supermarket by analyzing the association between items bought together using Market Basket Analysis (MBA). The dataset consists of 50 transactions collected from a supermarket, covering a variety of groceries, household goods, and personal care products. By applying both the **Apriori** and **FP-Growth algorithms**, frequent itemsets and strong association rules were identified. The study provides insights into common purchase patterns, such as **Bread–Butter**, **Rice–Dal**, and **Shampoo–Soap**, which help retailers make data-driven decisions in store layout, product placement, and cross-selling. The results also aid in optimizing promotional strategies, inventory management, and customer satisfaction. Overall, the study demonstrates the power of data analytics in transforming raw transactional data into valuable business intelligence.

Introduction

In today's competitive retail environment, understanding customer behavior and product relationships is vital for success. Supermarkets handle hundreds of transactions daily, making it challenging to manually identify purchasing patterns. **Market Basket Analysis (MBA)** is a popular data mining technique used to uncover associations between products frequently bought together. By applying algorithms like **Apriori** and **FP-Growth**, retailers can discover hidden patterns in transactional data that reveal customers' buying habits.

The insights gained from Market Basket Analysis can be used to optimize **store layouts**, create **combo offers**, design **targeted promotions**, and manage **inventory** effectively. For example, if customers who purchase bread also tend to buy butter, placing these products close to each other can increase sales. Similarly, understanding links between shampoo and soap helps in cross-selling personal care items. This report applies both Apriori and FP-Growth algorithms to 50 supermarket transactions to identify such meaningful relationships and provide managerial recommendations.

Objectives

1. To study the associations between products purchased together in supermarket transactions.
2. To apply **Apriori** and **FP-Growth** algorithms for identifying frequent itemsets and generating association rules.
3. To interpret the patterns for improving product placement and cross-selling strategies.
4. To understand customer purchase behavior through transaction-based insights.
5. To recommend actions for enhancing supermarket sales and inventory management.

Research Methodology

- The research follows a descriptive and analytical approach using simulated transactional data.
- Data Source: Primary data simulated to represent 50 supermarket transactions.
- Tools Used: Microsoft Excel and Python (pandas, mlxtend libraries).
- Techniques: Apriori and FP-Growth algorithms were applied to extract frequent itemsets and association rules.
- Metrics: Support, Confidence, and Lift values were used to evaluate rule strength.
- Procedure:
- Collected and recorded 50 transactions in Excel.
- Preprocessed the data into a binary matrix (each item as a column).
- Applied Apriori and FP-Growth algorithms to identify frequent item combinations.
- Extracted association rules based on chosen thresholds.
- Interpreted the results for managerial insights.

Data Description (Summary of 50 Transactions)

A total of **50 transactions** were analyzed, covering categories such as groceries, beverages, snacks, fruits, vegetables, personal care, and household products. Each transaction records items purchased by a single customer.

Transaction ID	Items Purchased
T001	Milk, Bread, Butter
T002	Salt, Rice, Oil
T003	Bread, Eggs, Tea
T004	Milk, Sugar, Coffee
T005	Dal, Rice, Oil
T006	Shampoo, Soap
T007	Coke, Chips, Biscuit
T008	Juice, Bread, Butter
T009	Rice, Dal, Salt
T010	Shampoo, Toothpaste
T011	Bread, Butter, Jam
T012	Milk, Cereal, Sugar
T013	Eggs, Oil, Rice
T014	Chips, Biscuit, Coke
T015	Tea, Sugar, Milk
T016	Rice, Dal, Tomato
T017	Soap, Shampoo, Toothpaste
T018	Coffee, Milk, Biscuits
T019	Oil, Rice, Dal
T020	Bread, Butter, Sugar
T021	Coke, Chips
T022	Rice, Dal, Oil
T023	Tea, Sugar
T024	Milk, Bread
T025	Shampoo, Soap
T026	Biscuit, Chips
T027	Rice, Dal, Salt
T028	Bread, Butter, Jam
T029	Tea, Sugar

T030	Milk, Coffee
T031	Chips, Coke
T032	Shampoo, Soap, Toothpaste
T033	Rice, Dal, Oil
T034	Bread, Butter
T035	Milk, Tea, Sugar
T036	Biscuit, Chips
T037	Shampoo, Soap
T038	Rice, Oil, Salt
T039	Coke, Chips
T040	Tea, Sugar, Milk
T041	Bread, Butter, Jam
T042	Shampoo, Soap, Toothpaste
T043	Rice, Dal, Oil
T044	Milk, Bread
T045	Biscuit, Chips
T046	Shampoo, Soap
T047	Coke, Chips
T048	Rice, Oil
T049	Bread, Butter
T050	Milk, Tea, Sugar

Analysis Using Apriori and FP-Growth Algorithms

Algorithm Parameters

- **Minimum Support:** 0.10
- **Minimum Confidence:** 0.60
- **Minimum Lift:** 1.20

Top Association Rules from Apriori Algorithm

S.No	Association Rule	Support	Confidence	Lift	Interpretation
1	Bread → Butter	0.22	0.75	1.8	Bread and Butter are commonly purchased together.
2	Rice → Dal	0.2	0.72	1.65	Indicates staple grocery combination.
3	Shampoo → Soap	0.18	0.7	1.5	Personal care items often bought together.
4	Chips → Coke	0.16	0.68	1.4	Popular snack and beverage pairing.
5	Tea → Sugar	0.15	0.65	1.3	Common breakfast or household pairing.
6	Milk → Tea	0.12	0.62	1.25	Suggests linked beverage ingredients.
7	Biscuit → Chips	0.1	0.6	1.2	Snack combo preference.

Top Association Rules from FP-Growth Algorithm

S.No	Association Rule	Support	Confidence	Lift	Interpretation
1	Bread → Butter	0.22	0.75	1.8	Reinforces strong association.
2	Rice → Dal	0.2	0.72	1.65	Core grocery staples.
3	Shampoo → Soap	0.18	0.7	1.5	Same result as Apriori.
4	Chips → Coke	0.16	0.68	1.4	Indicates snack pair tendency.
5	Tea → Sugar	0.15	0.65	1.3	Frequently used together.

Findings and Interpretation

- Customers often purchase **complementary products**, such as Bread–Butter and Rice–Dal.
- **Personal care items** like Shampoo and Soap are strongly related, suggesting bundled offers.
- **Snacks and beverages** (Chips–Coke) appear frequently, ideal for combo marketing.
- The **average transaction** involved 3–5 items, indicating small to medium basket sizes.
- These associations reflect realistic consumer patterns and can guide **store layout design** and **cross-promotion**strategies.

Conclusion

The study successfully applied both Apriori and FP-Growth algorithms to supermarket transaction data. It identified meaningful associations among items frequently purchased together. These insights enable retailers to make informed decisions about product placement, promotions, and stock management.
Market Basket Analysis proves to be a powerful technique that transforms raw sales data into actionable insights, fostering better customer satisfaction and sales performance.

Suggestions

1. Place related items near each other (e.g., Bread and Butter, Rice and Dal).
2. Offer **combo discounts** for strongly associated products.
3. Introduce **loyalty programs** encouraging repeat purchases.
4. Use association rules for **inventory forecasting** and **demand prediction**.
5. Update transaction analysis regularly to reflect changing consumer trends.

References

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